6G and Beyond

The Prospects for Intelligent Wireless Connectivity

完整排练稿｜所有成员衔接语加粗｜中文 PPT 页码提示

分工按“你讲第 1、4 章；队友 A 讲第 2、3 章；队友 B 讲第 5、6 章”安排。队友代号可替换为真实姓名。

读法：黑色英文为讲述内容；粗体英文为开场、交接、回接或章节衔接；蓝色微软雅黑中文为操作提示，不朗读。引用编号不朗读。

页码：所有中文标注均指 PPT 右下角实际页码，不是讲稿页码或 PPT 顶部栏目编号。特别注意：PPT 第 9 页顶部写着“07 / APPLICATIONS”；此版 PPT 含目录，共 14 页。

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朗读英文约 1203 词（含新增衔接语，不含提示、标题与参考文献）。作业要求全组 12 分钟，请计入停顿、切页和交接后排练。

本稿保留原讲稿的技术内容和参考文献；保留下午问候与全组称谓，加入目录讲述、衔接语和更新后的 PPT 页码提示。

1. Introduction

【本章讲述人：你】

【PPT 第 1 页｜封面｜你开场；英文读出，中文提示不读】

Good afternoon, everyone. Today, our group will present “6G and Beyond: The Prospects for Intelligent Wireless Connectivity.”

【切到 PPT 第 2 页｜新增目录｜仍由你介绍全组路线；随后切到第 3 页进入引言】

Here is our roadmap. We will first introduce 6G and its evolution, then cover its core technologies and applications. Finally, we will discuss challenges and future research directions before concluding.

【切到 PPT 第 3 页｜先讲左侧虚拟实验室场景，再对应右侧 CONNECT／SENSE／ADAPT】

Imagine attending a laboratory class from home, while interacting with a detailed virtual experiment and receiving immediate feedback. Now imagine the wireless network supporting that experience can also sense movement and adapt its resources as conditions change.

This is the broader ambition behind sixth-generation mobile networks, or 6G. The question is not simply how fast we can download a file. It is how communication, sensing, and intelligence can work together.

【仍在 PPT 第 3 页｜说明 IMT-2030 框架与文献综述性质；底部路线无需重复朗读】

This presentation compares research surveys with the International Telecommunication Union's IMT-2030 framework. It is a literature-based overview, rather than a report of a new experiment. [1], [2]

【交接：你 → 队友 A｜你说完下句后，由队友 A 切到 PPT 第 4 页并接讲】

I’ll now hand over to my teammate, who will explain the background and the technologies behind 6G.

2. Background and Technological Evolution

【本章讲述人：队友 A】

【PPT 第 4 页｜队友 A 接讲｜技术演进：从 1G 语音到 6G 融合系统】

Thank you.

To understand 6G, it helps to look at what earlier generations changed. First-generation networks supported analog voice calls. Second-generation networks introduced digital voice and messaging. Third-generation networks expanded mobile internet access, while fourth-generation networks made mobile broadband central to everyday life.

Fifth-generation networks broadened the focus beyond smartphones. Their main service categories include enhanced mobile broadband, highly reliable communication with low delay, and connections for large numbers of devices. [2]

Why investigate another generation? Future applications may require several capabilities together: detailed immersive content, rapid responses, accurate positioning, and dependable connectivity while users move.

【切到 PPT 第 5 页｜IMT-2030 框架；最后强调框架是发展目标】

The ITU's IMT-2030 framework reflects this broader ambition. Alongside improvements to communication, it includes artificial intelligence and communication, integrated sensing and communication, and ubiquitous connectivity. Sustainability and security are also central concerns. [1]

However, a framework is not a finished commercial system. We should treat proposed 6G capabilities as development goals, rather than assume every future connection will achieve every target simultaneously.

【下一章仍由队友 A 继续，无需换人】

3. Core Technologies of 6G

【本章讲述人：队友 A】

【PPT 第 6 页｜队友 A 继续｜四类核心技术总览】

With this framework in mind, let’s turn to the core technologies of 6G.

I will focus on four groups of technologies.

【第 6 页停留片刻指示四个技术模块，再切到 PPT 第 7 页｜AI-native networking 与 New radio approaches】

The first is AI-native networking. The idea is to design network functions with artificial intelligence in mind from the beginning. Learning algorithms could predict traffic, allocate resources, or help maintain connections as conditions change. Letaief and colleagues describe a two-way relationship: AI can improve wireless networks, and wireless networks can support distributed AI. Edge computing complements this approach by processing some data closer to users. However, AI also requires training data, computing resources, and reliable decisions. [3]

The second group concerns radio capacity and the propagation environment. Sub-terahertz and terahertz research explores wider bandwidths for demanding applications. Large antenna arrays can direct signals towards users, while reconfigurable intelligent surfaces can adjust how radio waves are reflected. These approaches are promising, but they face blockage, propagation loss, and hardware constraints. Higher frequencies would therefore complement lower-frequency coverage, rather than solve every connectivity problem. [2]

【切到 PPT 第 8 页｜ISAC 与 Integrated coverage：感知通信一体化、地面与非地面网络融合】

The third is integrated sensing and communication, or ISAC. A wireless system would transmit information while also using radio signals to estimate properties such as an object's location or movement. Liu and colleagues explain that sharing signals and infrastructure can bring benefits, but sensing accuracy and communication performance may compete for resources. [4]

Finally, researchers are investigating closer integration between terrestrial networks, satellites, and aerial platforms. This could extend connectivity beyond conventional ground infrastructure. The difficulty is coordinating very different links, especially when satellites, aircraft, and users are moving. [2]

【交接：队友 A → 你｜队友 A 说完下句后，由你切到 PPT 第 9 页】

With these technologies in mind, I’ll hand back to my teammate to discuss three potential applications.

4. Applications and Case Studies

【本章讲述人：你】

【PPT 第 9 页｜你再次接讲｜注意：顶部写 07 / APPLICATIONS，右下角实际页码为 09】

Thank you.

What could these technologies mean in practice? I will use three literature-based application cases. They illustrate proposed designs, not completed commercial 6G deployments.

【仍在 PPT 第 9 页｜左侧 CASE 01：沉浸式教育；需求 → 技术帮助 → 评价指标】

The first is immersive education. Consider students sharing a virtual laboratory. High-quality images require substantial capacity, while interaction requires consistent response times. Edge computing could support nearby rendering and processing, and intelligent resource allocation could respond to changing demand. Research on extended reality also reminds us that network performance should be assessed through human experience, not only data rates. A useful evaluation would measure interaction delay, interruptions, and user comfort. [2], [5]

【仍在 PPT 第 9 页｜中间 CASE 02：互联交通；handover 指车辆移动时的网络／基站切换】

The second case is connected transport. In the ISAC survey, Liu and colleagues discuss roadside infrastructure that can sense a vehicle's position and motion while supporting communication. Sensing information can help track a narrow radio beam or prepare resources before the vehicle enters another coverage area. This connects sensing directly to a communication problem. However, tracking errors and interference can undermine the benefit. Evaluation should therefore examine both sensing accuracy and connection reliability. [4]

【仍在 PPT 第 9 页｜右侧 CASE 03：可再生能源协调；供需信息、成本、能耗、兼容性】

The third case is renewable-energy coordination. Rashid and colleagues review how future communication networks could support energy management, distributed generation, and electricity trading. For example, timely information from solar installations, batteries, and consumers could help coordinate local supply and demand. Yet the authors also identify costs, power consumption, and compatibility with older equipment as obstacles. [6]

【仍在 PPT 第 9 页｜对应底部蓝色问题：比现有方案带来什么可测量收益？】

These cases suggest an important test: what extra benefit does a proposed 6G design provide compared with an existing solution? A new network should be justified by measurable improvements, not by the generation label alone.

【交接：你 → 队友 B｜你说完下句后，由队友 B 切到 PPT 第 10 页】

I’ll now hand over to our final speaker to discuss the challenges and future research directions.

5. Challenges and Future Research Directions

【本章讲述人：队友 B】

【PPT 第 10 页｜队友 B 接讲｜依次对应 SYSTEM、TRUST、IMPACT 三个框】

Thank you.

Three challenges deserve particular attention.

First, promising components must work together under realistic conditions. A high-capacity link may be sensitive to blockage. A complex AI model may consume more energy than its decisions save. Research should compare complete systems under mobility, congestion, and equipment constraints. Reproducible testbeds and fair baseline comparisons would help show where the benefits actually come from. [2], [3]

Second, intelligence and sensing create security and privacy risks. A network that learns from data can also learn from manipulated data. A network that senses movement may reveal sensitive information about people. Nguyen and colleagues review threats across prospective 6G technologies, including attacks on machine-learning processes. Future research should build protection into system design and test how networks behave when data or models are attacked. [7]

Third, sustainability requires more than transmitting each bit efficiently. Additional infrastructure, computing, and hardware also consume resources. Ahmad and colleagues examine sustainability across network management, security, and hardware. Their analysis supports evaluating resource trade-offs across the system. Future studies should report energy consumption alongside performance and consider the practical cost of deploying and maintaining their designs. [8]

【切到 PPT 第 11 页｜三个研究问题：可靠性、抵抗攻击与恢复能力、可持续性】

For these reasons, I would prioritise three research questions: Does the system remain reliable outside ideal conditions? Can it protect users and recover from attacks? And does its benefit justify its energy and deployment costs?

Answering these questions will require collaboration across communications, computing, security, and application design.

【下一章仍由队友 B 继续，无需换人】

6. Conclusion

【本章讲述人：队友 B】

【切到 PPT 第 12 页｜队友 B 继续｜结论；下句是同一人的章节衔接】

Let me finish with our main message.

To conclude, 6G research points towards wireless systems that combine communication with sensing, computing, and intelligence. AI-native networking, new radio techniques, ISAC, and broader coverage could support applications ranging from immersive learning to connected transport and energy coordination.

But these opportunities come with difficult trade-offs. Faster transmission alone does not guarantee a dependable application, and a more intelligent network is not automatically secure or sustainable.

The most convincing future for 6G is therefore one demonstrated through useful applications and realistic evidence: better experiences, reliable operation, and responsible use of resources.

【切到 PPT 第 13 页｜参考文献与致谢；第 14 页为其余参考文献，按需要展示，不逐条朗读】

Our references are shown on the final two slides. Thank you for listening. We welcome your questions.

References

【PPT 第 13 页列[1]–[4]；第 14 页列[5]–[8]。以下为原讲稿参考文献，供查阅，不朗读。】

[1] International Telecommunication Union, “Framework and overall objectives of the future development of IMT for 2030 and beyond,” Recommendation ITU-R M.2160-0, Nov. 2023. [Online]. Available: https://www.itu.int/rec/R-REC-M.2160-0-202311-I/en. Accessed: Oct. 21, 2026.

[2] S. Latreche and H. Bellahsene, “A comprehensive survey on 6G: Enabling technologies, key applications, and future challenges,” Franklin Open, vol. 15, Art. no. 100559, 2026, doi: 10.1016/j.fraope.2026.100559. https://doi.org/10.1016/j.fraope.2026.100559

[3] K. B. Letaief, W. Chen, Y. Shi, J. Zhang, and Y.-J. A. Zhang, “The roadmap to 6G – AI empowered wireless networks,” arXiv:1904.11686v2, 2019. [Online]. Available: https://arxiv.org/abs/1904.11686v2

[4] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, “Integrated sensing and communications: Towards dual-functional wireless networks for 6G and beyond,” arXiv:2108.07165v1, 2021. [Online]. Available: https://arxiv.org/abs/2108.07165v1

[5] W. Saad, M. Bennis, and M. Chen, “A vision of 6G wireless systems: Applications, trends, technologies, and open research problems,” arXiv:1902.10265v1, 2019. [Online]. Available: https://arxiv.org/abs/1902.10265v1

[6] A. Rashid, P. Biswas, A. K. M. A. Habib, A. F. M. Z. Abadin, A. Ghosh, and S. M. A. Motakabber, “Review on 6G communication network technology for sustainable renewable energy system: Implementations, constraints, and recommendations,” Results in Engineering, vol. 28, Art. no. 107933, 2025, doi: 10.1016/j.rineng.2025.107933. https://doi.org/10.1016/j.rineng.2025.107933

[7] V.-L. Nguyen, P.-C. Lin, B.-C. Cheng, R.-H. Hwang, and Y.-D. Lin, “Security and privacy for 6G: A survey on prospective technologies and challenges,” arXiv:2108.11861v2, 2021. [Online]. Available: https://arxiv.org/abs/2108.11861v2

[8] I. Ahmad et al., “Sustainability in 6G networks: Vision and directions,” in Proc. 2023 IEEE Conf. Standards for Communications and Networking (CSCN), pp. 202–208, doi: 10.1109/CSCN60443.2023.10453200. https://doi.org/10.1109/CSCN60443.2023.10453200